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Ram air turbine powered cabin air compressor

Abstract

A pressurization system for a vehicle includes an inlet for receiving a medium and a cabin air compressing device arranged in fluid communication with the inlet. The cabin air compressing device includes a shaft, a compressor connected to the shaft, a motor operably connected to the shaft and configured to drive the compressor, and a turbine connected to the shaft and also configured to drive the compressor upon receipt of the medium. A heat exchanger has a first inlet fluidly connected to the compressor outlet and a second inlet fluidly connected to the turbine outlet. The compressor inlet and the turbine inlet are fluidly connected to and are arranged in parallel relative to the inlet.

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Background/Summary

BACKGROUND

- (1) Exemplary embodiments pertain to the art of thermal management of and, in particular, to thermal management of one or more loads of a vehicle.
- (2) A typical commercial aircraft includes at least several nonintegrated pressurization systems configured to provide temperature control to various regions of the aircraft. For example, an aircraft environmental control system primarily provides heating and cooling for the aircraft cabin area. In addition, a galley chiller system is dedicated to refrigerating the food carts in the galleys located throughout the aircraft. Since each system has a significant weight and power requirement, the overall efficiency of the aircraft is affected by these nonintegrated systems.
- (3) One or more of these pressurization systems may rely on ram or fresh air to condition, i.e., to cool or heat another medium. However, in applications where the aircraft is travelling at supersonic speeds, the temperature of the ram air may be too high to effectively remove heat from another load.

BRIEF DESCRIPTION

- (4) According to an embodiment, a pressurization system for a vehicle includes an inlet for receiving a medium and a cabin air compressing device arranged in fluid communication with the inlet. The cabin air compressing device includes a shaft, a compressor connected to the shaft, a motor operably connected to the shaft and configured to drive the compressor, and a turbine connected to the shaft and also configured to drive the compressor upon receipt of the medium. A heat exchanger has a first inlet fluidly connected to the compressor outlet and a second inlet fluidly connected to the turbine outlet. The compressor inlet and the turbine inlet are fluidly connected to and are arranged in parallel relative to the inlet.
- (5) In addition to one or more of the features described herein, or as an alternative, in further embodiments the medium is ram air.
- (6) In addition to one or more of the features described herein, or as an alternative, in further embodiments an expanded medium is provided at the turbine outlet and a compressed medium is provided at the compressor outlet, and within the heat exchanger, heat is configured to transfer from the compressed medium to the expanded medium.
- (7) In addition to one or more of the features described herein, or as an alternative, in further embodiments the compressed medium at an outlet of the heat exchanger is provided to at least one component configured to further condition the compressed medium.
- (8) In addition to one or more of the features described herein, or as an alternative, in further embodiments the at least one component configured to further condition the compressed medium further comprises a load heat exchanger.
- (9) In addition to one or more of the features described herein, or as an alternative, in further embodiments the expanded medium provided at an outlet of the heat exchanger is exhausted overboard.
- (10) In addition to one or more of the features described herein, or as an alternative, in further embodiments the expanded medium at the outlet of the heat exchanger is provided to another cooling heat exchanger.
- (11) In addition to one or more of the features described herein, or as an alternative, in further embodiments comprising a bypass conduit connecting the inlet and the second inlet of the heat exchanger and a valve associated with the bypass conduit.
- (12) In addition to one or more of the features described herein, or as an alternative, in further embodiments the vehicle is an aircraft.
- (13) According to an embodiment, a method of conditioning a load of a vehicle includes compressing a first portion of a medium provided from an inlet within a compressor to create a compressed medium that is output at a compressor outlet, expanding a second portion of the medium provided from the inlet within a turbine to create an expanded medium and cooling the compressed medium within a cooling heat exchanger connected to the compressor outlet via a cooling medium. Compressing the first portion of the medium and expanding the second portion of the medium occur simultaneously.
- (14) In addition to one or more of the features described herein, or as an alternative, in further embodiments the expanded medium output from the turbine is provided to the cooling heat exchanger as the cooling medium.
- (15) In addition to one or more of the features described herein, or as an alternative, in further embodiments in a first mode, the compressor is driven in response to expanding the second portion of the medium within the turbine, and in a second mode, the compressor is driven by an electric motor and in response to expanding the second portion of the medium within the turbine.
- (16) In addition to one or more of the features described herein, or as an alternative, in further embodiments exhausting the cooling medium from the cooling heat exchanger overboard.
- (17) In addition to one or more of the features described herein, or as an alternative, in further embodiments providing the cooling medium from the cooling heat exchanger to another cooling heat exchanger.

- (18) In addition to one or more of the features described herein, or as an alternative, in further embodiments delivering the compressed medium provided at an outlet of the cooling heat exchanger to at least one component configured to further condition the compressed medium.
- (19) In addition to one or more of the features described herein, or as an alternative, in further embodiments further cooling the expanded medium within a load heat exchanger via another cooling medium.
- (20) In addition to one or more of the features described herein, or as an alternative, in further embodiments opening a valve arranged within a bypass conduit such that at least some of the second portion of the medium provided from the inlet bypasses the turbine.
- (21) In addition to one or more of the features described herein, or as an alternative, in further embodiments the medium is ram air.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is a schematic diagram of a portion of a pressurization system of a vehicle according to an embodiment; and
- (3) FIG. 2 is a schematic diagram of a pressurization system of a vehicle according to an embodiment.

DETAILED DESCRIPTION

- (4) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
- (5) With reference now to FIG. 1, a schematic diagram of a portion of a pressurization system **20** is illustrated. The pressurization system **20** may be a refrigeration or air cycle subsystem and is configured to receive a medium A at an inlet **22**. In the illustrated, non-limiting embodiment, the medium is fresh air, such as outside air for example. This outside air, also referred to herein as RAM air, can be procured via one or more scooping mechanisms, such as an impact scoop or a flush scoop for example. When the pressurization system **20** is implemented on an aircraft, the medium A is generally at an ambient pressure equal to an air pressure outside of the aircraft when the aircraft is on the ground and is between an ambient pressure and a cabin pressure when the aircraft is in flight. It should be understood that any suitable medium A is within the scope of the disclosure. For example, other suitable mediums available on an aircraft include, but are not limited to bleed air, which is pressurized air originating from, i.e., being “bled” from, an engine or auxiliary power unit of the aircraft, or cabin discharge air, which is air leaving the cabin and that would typically be discharged overboard.
- (6) It should be understood that the elements of the pressurization system **20** are connected via valves, tubes, pipes, and the like. Valves (e.g., flow regulation device or mass flow valve) are devices that regulate, direct, and/or control a flow of a medium by opening, closing, or partially obstructing various passageways within the tubes, pipes, etc. of the system. Valves can be operated by actuators, such that flow rates of the medium in any portion of the pressurization system **20** can be regulated to a desired value.
- (7) In the illustrated, non-limiting embodiment, the pressurization system includes at least one cabin air compressing device **24**. The cabin air compressing device **24** is a mechanical device that includes one or more components for performing thermodynamic work on a medium (e.g., extracts work from the medium by raising and/or lowering the pressure thereof and by raising and/or lowering the temperature thereof).
- (8) In the illustrated, non-limiting embodiment, the cabin air compressing device **24** includes a

compressor **26** and a turbine **28** operably coupled by a shaft **30**. Accordingly, the cabin air compressing device **24** may also be referred to herein as a turboCAC. A compressor **26** is a mechanical device configured to raise a pressure of a medium and can be driven by another mechanical device (e.g., a motor or a medium via a turbine). Examples of compressor types include centrifugal, diagonal or mixed-flow, axial-flow, reciprocating, ionic liquid piston, rotary screw, rotary vane, scroll, diaphragm, air bubble, etc. A turbine **28** is a mechanical device that expands a medium and extracts work therefrom (also referred to as extracting energy) to drive the compressor **26** via the shaft **30**. Although only a two-wheel cabin air compressing device **24** is illustrated and described herein, it should be understood that embodiments having additional wheels, such as an additional compressor, turbine, and/or fan for example, are also within the scope of the disclosure.

(9) In the illustrated, non-limiting embodiment, an inlet **32** of the compressor **26** and an inlet **34** of the turbine **28** are both fluidly connected to the inlet **22**. Accordingly, a first portion of the flow of medium A from the inlet **22** may be provided to the compressor inlet **32** and a second portion of the flow of medium A from the inlet **22** may be provided to the turbine inlet **34** in parallel.

(10) In an embodiment, the cabin air compressing device **24** additionally includes an electric motor **36** connected to the shaft **30** and operable to drive the compressor **26**. The motor **36** can receive power from a power source (not shown) such as generator or a power bus (e.g., a power bus of an aircraft). In embodiments including both a turbine **28** and a motor **36** connected to the shaft **30**, either or all of the turbine **28** and the motor **36** may be used to drive the compressor **26**.

(11) The pressurization system **20** includes at least one cooling heat exchanger **38** arranged downstream from at least one of the compressor **26** and the turbine **28** relative to a flow of medium through the pressurization system **20**. In the illustrated, non-limiting embodiment, the cooling heat exchanger **38** is configured to receive and cool a flow of compressed medium output from the compressor **26**. As shown, an outlet **40** of the compressor **26** is connected to a first or heated flow inlet **42** of the heat exchanger via a conduit **44**. The compressed medium A1 is cooled within the heat exchanger **38** by a cooling medium provided at a second or cooling flow inlet **46**. In the illustrated, non-limiting embodiment, the cooling medium is the expanded medium A2 provided at the outlet **48** of the turbine **28**, the outlet **48** being connected to the cooling flow inlet **46** via a conduit **50**. Although the heat exchanger **38** is illustrated in FIG. 1 as a cross-flow heat exchanger, a skilled artisan will realize that the heat exchanger **38** can be any suitable type of heat exchanger that achieves the desired result of cooling the compressed medium A1.

(12) The cooling heat exchanger **38** additionally includes a first or heated flow outlet **52** and a second or cooling flow outlet **54**. In an embodiment, the first heated flow outlet **52** of the heat exchanger **38** is fluidly coupled to a volume of the vehicle, such as a cabin for example. The first heated flow outlet **52** may be directly connected to a volume, such as via a conduit, or alternatively, the flow at the first heated flow outlet **52** may be provided to at least one component, such as of an environmental control system (ECS), operable to condition the compressed medium A1 for example, before it is delivered to the volume. The heated expanded medium A2 provided at the second cooling flow outlet **54** is exhausted overboard to the ambient atmosphere surrounding the vehicle. In an embodiment, the expanded medium A2 may be exhausted overboard directly from the second cooling flow outlet **54**, or may be provided to one or more other components before ultimately being exhausted overboard.

(13) During operation of the pressurization system **20** in a first mode when the vehicle is at a high altitude, such as during a supersonic cruise condition for example, a first portion of the medium A provided at the inlet **22** is provided generally directly to the inlet **32** of the compressor **26**. The act of compressing the medium A, heats the medium A and increases the pressure of the medium A. At the same time, a second portion of the flow of medium A from the inlet **22** is provided to the inlet **34** of the turbine **28**. Within the turbine **28**, the medium A is expanded and work is extracted therefrom. The work extracted from the medium A within the turbine **28** is used to drive the compressor **26**. However, it should be understood that embodiments where the aircraft is not flying

in a supersonic condition, the motor **36** may be used in combination with the energy extracted at the turbine to drive the compressor **26**. The expanded medium **A2** output from the turbine **28** has a reduced temperature and pressure relative to the medium **A** provided to the inlet **34** of the turbine **28**.

(14) The compressed medium **A1** output from the outlet **40** of the compressor **26**, represented as **A1**, may then flow to the heated flow inlet **42** of the cooling heat exchanger **38**. Similarly, the expanded medium output from the turbine **28**, represented as **A2**, may then flow to the cooling flow inlet **46** of the cooling heat exchanger **38**. Within the cooling heat exchanger **38**, heat from the compressed medium **A1** is transferred to the cooling medium **A2**, before the cooling medium **A2** is exhausted overboard, such as via the second cooling flow outlet **54** for example, into the ambient atmosphere adjacent to the vehicle. As previously described, the compressed medium **A1** provided at the heated flow outlet **52** of the cooling heat exchanger **38** is then provided to the volume or to one or more other components arranged upstream from the volume and configured to further condition the compressed medium **A1**.

(15) A second mode of operation of the pressurization system **20** is substantially similar to the first mode of operation. However, in the second mode, a valve **V1** arranged within a bypass conduit **56** is opened such that at least some, and in some embodiments all of the second portion of the flow **A** bypasses the turbine **28** via the bypass conduit **56**. In such embodiments, the cooling medium provided to the second cooling flow inlet **46** of the cooling heat exchanger **38** is either the medium **A** or a combination of the expanded medium **A2** and the medium **A**.

(16) With reference now to FIG. 2, an example of a pressurization system **20** is provided. In the illustrated, non-limiting embodiment, the pressurization system **20** includes a first cabin air compressing device **24a** and a second cabin air compressing device **24b**, each of which includes a compressor **26a**, **26b**, turbine **28a**, **28b**, and motor **36a**, **36b** operably coupled to a shaft **30a**, **30b** as previously described. The first compressor **26a** and the first turbine **28a** are arranged in parallel relative to the inlet **22** as previously described. In addition, the second cabin air compressing device **24b** is arranged in parallel with a portion of the first cabin air compressing device **24a**. In the illustrated, non-limiting embodiment, the second compressor **26b** is arranged in parallel with the first compressor **26a** and the first turbine **28a** relative to a flow of medium **A** provided at the inlet **22**. Further, in an embodiment, the medium **A** provided at the inlet **22** is configured to flow through compressor **26b** and the turbine **28b** of the second cabin air compressing device **24b** in series, as will be described in more detail below.

(17) In the non-limiting embodiment of FIG. 2, the at least one cooling heat exchanger includes both a first cooling heat exchanger **38** and another cooling heat exchanger or second cooling heat exchanger **60** (also referred to herein as an outflow or regeneration heat exchanger). In the illustrated, non-limiting embodiment, these cooling heat exchangers **38**, **60** are co-located; however, this is not required. In one embodiment, the cooling heat exchangers **38**, **60** are contained in a single unit. For simplicity, in FIG. 2, the connection that provides the transfer of the cooling medium from the first cooling heat exchanger **38** to the second heat exchanger **60** is not shown. As such, the unit comprising the first cooling heat exchanger **38** and the second cooling heat exchanger **60** includes a cooling medium inlet **46** at the first cooling heat exchanger **38** and cooling medium outlet **62** that is part of second cooling heat exchanger **60**. Although not shown, the first cooling heat exchanger **38** may include a cooling medium outlet (not shown) that provides the flow of cooling medium directly (or indirectly) to a cooling flow inlet (not shown) of the second cooling heat exchanger **60**. It shall be understood that the input/output need not be a physical object and can be formed such that the path from the heated flow inlet **46** to the outlet **62** is a continuous path.

(18) In the illustrated, non-limiting embodiment, the pressurization system **20** additionally includes at least one heat exchanger arranged downstream from one or more components of the second cabin air compression device **24b**. In the non-limiting embodiment of FIGS. 1 and 2, the at least one heat exchanger includes a first heat exchanger **70**, also referred to herein as a load heat

exchanger, and a second heat exchanger **72**. These heat exchangers **70**, **72** may, but need not be contained in a single unit. The first heat exchanger **70** includes a first inlet **74** and a first outlet **76** arranged along a flow path and the second heat exchanger **72** similarly includes a first inlet **80** and a first outlet **82** arranged along a flow path. As shown, the first inlet **74** of the first heat exchanger **70** may be fluidly connected, directly or indirectly, to the heated flow outlet **52** of the first cooling heat exchanger **38**. Alternatively, or in addition, the first inlet **80** of the second heat exchanger **72** may be fluidly connected directly, or indirectly, to a first or heated flow outlet **64** of the second cooling heat exchanger **60**.

(19) The unit comprising the first heat exchanger **70** and the second heat exchanger **72** includes a second or cooling medium inlet **78** at the first heat exchanger **70** and second or cooling medium outlet **84** that is part of second heat exchanger **72**. Although not shown, the first heat exchanger **70** may include a cooling medium outlet (not shown) that provides the flow of cooling medium directly (or indirectly) to a cooling flow inlet (not shown) of the second heat exchanger **72** to define a secondary cooling flow path through the heat exchangers **70**, **72**. It shall be understood that the input/output need not be a physical object and can be formed such that the path from the second cooling medium inlet **78** to the second cooling medium outlet **84** is a continuous path.

(20) Another cooling medium, also referred to herein as a second cooling medium, is provided to the cooling flow path extending through between the second cooling medium inlet **78** to the second cooling medium outlet **84** of the heat exchangers **70**, **72**. In the illustrated, non-limiting embodiment, the compressed medium output from the compressors **24a** and **24b** are cooled within the first heat exchanger **70** and the second heat exchanger **72**, respectively, is cooled by a second cooling medium **A2b** provided at second cooling medium inlet **78**. In the illustrated, non-limiting embodiment, the second cooling medium is expanded medium output from the outlet of the turbine **28b** of the second cabin air compression device **24b**. Although the first and second cooling mediums are illustrated and described herein as being exhausted overboard to the ambient atmosphere surrounding the vehicle, it should be understood that in other embodiments, the cooling mediums may be provided to another load of the vehicle.

(21) In an embodiment, the pressurization system **20** additionally includes a turbofan **90** including a turbine **92** operably coupled to a fan **94** by a shaft **96**. As shown, the turbine **92** may be arranged downstream from and is in fluid communication with the first outlet **76** of the load heat exchanger **70** relative to a flow of compressed medium. The flow provided at the outlet of the turbine **92** may be further conditioned within a heat exchanger before being delivered to a load, such as the cabin for example.

(22) During operation of the pressurization system **20** of FIG. 2, the medium A provided at the inlet **22** is provided generally directly and simultaneously to each of an inlet of the compressor **26a**, the turbine **28a**, and the compressor **26b**. The medium A is compressed within the compressor **26a**, and as a result, the compressed medium **A1a** output from the compressor **26a** has an increased pressure and temperature relative to the medium A at the inlet **22**. Within the turbine **28a**, the medium A is expanded and work is extracted therefrom. The work extracted from the medium A within the turbine **28a** in combination with the energy provided by the motor **36a** is used to drive the compressor **26a**. The expanded medium **A2a** output from the turbine **28a** has a reduced temperature and pressure relative to the medium A provided to the inlet of the turbine **28a**. Similarly, the compressed medium **A1b** output from the compressor **26b** has an increased pressure and temperature relative to the medium A at the inlet **22**.

(23) The flow of first compressed medium **A1a** output from the compressor **26a** is provided to the heated flow inlet **44** of the first cooling heat exchanger **38**, and the flow of the second compressed medium **A1b** output from the compressor **26b** is provided to the heated flow inlet **66** of the second cooling heat exchanger **60**. Within each of the first cooling heat exchanger **38** and the second cooling heat exchanger **60**, the respective compressed mediums **A1a**, **A1b** are cooled by the first cooling medium. In the illustrated, non-limiting embodiment, the first cooling medium is the

expanded medium **A2a** output from the turbine **28a** of the first cabin air compression device **24a**. The first cooling medium **A2a** provided at the cooling flow outlet **62** of the second cooling heat exchanger **60** may be drawn through the fan **94** of the turbofan **90** and exhausted overboard. However, in other embodiments, the first cooling medium **A2a** provided at the cooling flow outlet **62** may be configured to bypass the fan **94** and is exhausted overboard.

(24) The compressed medium **A1a** output from the first cooling heat exchanger **38** is configured to pass through a flow path extending between the first inlet **74** and the first outlet **76** of the first heat exchanger **70**. At the same time, the compressed medium **A1b** output from the second cooling heat exchanger is configured to pass through a flow path extending between the first inlet **80** and the first outlet **82** of the second heat exchanger **72**. Within the second heat exchanger **72**, the compressed medium **A1b** is cooled by a second cooling medium, to be described in more detail below. From the first outlet **82** of the second heat exchanger **72**, the further cooled compressed medium **A1b** is provided to the inlet of the turbine **28b** of the second cabin air compression device **24b**.

(25) Within the turbine **28b**, the compressed medium **A1b** is expanded and work is extracted therefrom. The work extracted from the medium **A1b** within the turbine **28b** in combination with the energy provided by the motor **36b** is used to drive the compressor **26b**. The expanded medium **A2b** output from the turbine **28b** has a reduced temperature and pressure relative to the second compressed medium **A1b** provided to the inlet of the turbine **28b**.

(26) The expanded medium **A2b** output from the turbine **28b** is provided to the continuous fluid flow path extending between the second cooling medium inlet **78** of the first heat exchanger **70** and the second cooling medium outlet **84** of the second heat exchanger **72**. Accordingly, in an embodiment, the expanded medium **A2b** is heated via the heat exchange relationship with the compressed medium **A1a** in the first heat exchanger **70** and the heat exchange relationship with the compressed medium **A1b** in the second heat exchanger **72**. The resulting heated cooling medium **A2b** provided at the second outlet **82** may be exhausted overboard.

(27) In an embodiment, the cooled compressed medium **A1a** provided at the first outlet **76** of the first heat exchanger **70** passes through a water collector **98** before being provided to the inlet of the turbine **92** of the turbofan **90**. Within the turbine **92**, the compressed medium **A1a** is expanded and work is extracted therefrom. The work extracted from the medium **A1a** is used to drive the fan **94** via the shaft **96**. From the outlet of the turbine **92**, the expanded medium **A3** is provided to the cabin **100** or to one or more other loads. It should be understood that the configuration of the pressurization system **20** illustrated and described in FIG. 2 is intended as an example only and that any suitable configuration operable to condition one or more flows of medium therein is within the scope of the disclosure.

(28) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(29) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(30) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular

embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A pressurization system for a vehicle comprising: an inlet for receiving a medium; a cabin air compressing device arranged in fluid communication with the inlet, the cabin air compressing device comprising: a shaft; a compressor connected to the shaft, the compressor having a compressor inlet and a compressor outlet, the compressor inlet being fluidly connected to the inlet by a first conduit; an electric motor operably connected to the shaft and configured to drive the compressor; and a turbine connected to the shaft and also configured to drive the compressor upon receipt of the medium, the turbine having a turbine inlet and a turbine outlet, the turbine inlet being fluidly connected to the first conduit downstream from the inlet and upstream from the compressor inlet via a second conduit; and a heat exchanger having a first inlet fluidly connected to the compressor outlet and having a second inlet fluidly connected to the turbine outlet, the first inlet of the heat exchanger being directly connected to the compressor outlet; a bypass conduit having an inlet end fluidly connected to the second conduit at a location upstream from the turbine and an outlet end fluidly connected to the second inlet of the heat exchanger, wherein only one or more of a flow of medium output from the turbine and a flow of medium from the bypass conduit is provided to the second inlet of the heat exchanger; wherein the compressor inlet, the turbine inlet, and the bypass conduit are fluidly connected to and are arranged in parallel relative to the inlet.
2. The pressurization system of claim 1, wherein the medium is ram air.
3. The pressurization system of claim 1, wherein an expanded medium is provided at the turbine outlet and a compressed medium is provided at the compressor outlet, and within the heat exchanger, heat is configured to transfer from the compressed medium to the expanded medium.
4. The pressurization system of claim 3, wherein the compressed medium at an outlet of the heat exchanger is provided to at least one component configured to further condition the compressed medium.
5. The pressurization system of claim 4, wherein the at least one component configured to further condition the compressed medium further comprises a load heat exchanger.
6. The pressurization system of claim 3, wherein the expanded medium provided at an outlet of the heat exchanger is exhausted overboard.
7. The pressurization system of claim 6, wherein the expanded medium at the outlet of the heat exchanger is provided to another cooling heat exchanger.
8. The pressurization system of claim 1, further comprising: a valve associated with the bypass conduit.
9. The pressurization system of claim 1, wherein the vehicle is an aircraft.
10. A method of conditioning a load of a vehicle comprising: compressing a first portion of a medium provided from an inlet within a compressor to create a compressed medium that is output at a compressor outlet; expanding a second portion of the medium provided from the inlet within a turbine to create an expanded medium, wherein compressing the first portion of the medium and expanding the second portion of the medium occur simultaneously; providing the compressed medium from the compressor outlet directly to a cooling heat exchanger; cooling the compressed medium within the cooling heat exchanger connected to the compressor outlet via a cooling medium; operating a valve arranged within a bypass conduit such that at least some of the second portion of the medium provided from the inlet bypasses the turbine, wherein the cooling medium provided to the cooling heat exchanger is only one or more of the expanded medium and the at least some of the second portion of the medium that bypasses the turbine.
11. The method of claim 10, wherein the expanded medium output from the turbine is provided to the cooling heat exchanger as the cooling medium.

12. The method of claim 10, wherein in a first mode, the compressor is driven in response to expanding the second portion of the medium within the turbine, and in a second mode, the compressor is driven by an electric motor and in response to expanding the second portion of the medium within the turbine.
 13. The method of claim 10, further comprising exhausting the cooling medium from the cooling heat exchanger overboard.
 14. The method of claim 10, further comprising providing the cooling medium from the cooling heat exchanger to another cooling heat exchanger.
 15. The method of claim 10, further comprising delivering the compressed medium provided at an outlet of the cooling heat exchanger to at least one component configured to further condition the compressed medium.
 16. The method of claim 10, further comprising further cooling the expanded medium within a load heat exchanger via another cooling medium.
 17. The method of claim 10, wherein the medium is ram air.
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